

Connectedness

1.1 Connectedness

Definition 1.1.1

A topological space X is **connected** if and only if one of the following equivalent conditions holds:

- (i) X cannot be expressed as the union of two disjoint nonempty open sets.
- (ii) The subsets of X that are clopen in X are the empty set and X itself.
- (iii) Every continuous function $f : X \rightarrow \{0, 1\}$ is constant, where $\{0, 1\}$ is equipped with the discrete topology.

Proof. If (i) holds then since

$$X = f^{-1}(0) \cup f^{-1}(1)$$

one of the component must be empty. Thus f is constant. Then (ii) holds.

If (ii) holds, then for each U, V such that $X = U \cup V$, $U \cap V = \emptyset$, define

$$f := \begin{cases} 1, & x \in U \\ 0, & x \in V \end{cases}$$

then one of the U and V must be empty. □

Lemma 1.1.2

If Y is a subspace of X , a separation of Y is a pair of disjoint nonempty sets A and B whose union is Y , neither of which contains a limit point of the other.

The space Y is connected if there exists no separation of Y .

Proof. Suppose that A and B form a separation of Y . Then A is clopen in Y , $\bar{A} \cap Y = A$, equivalently, $\bar{A} \cap B = \emptyset$. It follows that B contains no limit point of A .

Conversely, suppose that A and B are disjoint nonempty sets whose union is Y , neither of which contains a limit point of the other. Then $\bar{A} \cap B = \emptyset$, $\bar{A} \cap Y = A$, A is closed in Y . Similarly, B is closed in Y , then A, B are clopen in Y . $\square$

Lemma 1.1.3

If the sets C and D form a separation of X , and if Y is a connected subspace of X , then Y lies entirely within either C or D

Proof. X is not connected, there is a continuous function $f : X \rightarrow \{0, 1\}$ with $f(C) = \{0\}$, $f(D) = \{1\}$. Then $f|_Y : Y \rightarrow \{0, 1\}$ is a continuous function. Since Y is connected, we have $f|_Y$ is constant. Assume that $f|_Y \equiv 0$, then $Y \subseteq f^{-1}(0) = C$. $\square$

Theorem 1.1.4

Let A be a connected subspace of X . If $A \subseteq B \subseteq \bar{A}$, then B is also connected.

Remark.

If B is formed by adjoining to the connected subspace A some or all of its limit points, then B is connected.

Proof. Let $f : B \rightarrow \{0, 1\}$ be continuous. Then f is constant on A since A is connected, we assume that $f(A) = \{0\}$. If $f(\bar{A}) \neq \{0\}$, then there exists $p \in \bar{A}$ such that $f(p) = 1$. Then $f^{-1}(1)$ is an open set containing p , thus $f^{-1}(1) \cap A = \emptyset$, a contradiction. $\square$

Definition 1.1.5

A topological space X is said to be **path connected** if and only if for all $x, y \in X$ there is a **path** that connects x and y .

Theorem 1.1.6

If X is (path) connected and $f : X \rightarrow Y$, then fX is (path) connected.

Corollary 1.1.7

Connected and **path connected** are topological properties.

Corollary 1.1.8

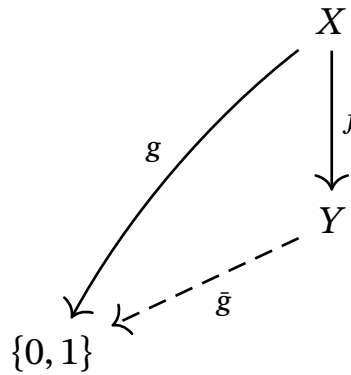
The quotient of a (path) connected space is (path) connected.

Proof. Since quotient maps are continuous surjections, we know quotients preserve (path) connectedness. $\square$

Theorem 1.1.9

Let X be a space and $f : X \rightarrow Y$ be a surjective map. If Y is connected in the quotient topology and if each fiber $f^{-1}y$ is connected, then X is connected.

Proof. Let $g : X \rightarrow \{0, 1\}$. Since the fibers of f are connected, g must be constant on each fiber of f . Therefore g factors through $f : X \rightarrow Y$, and there is a map $\bar{g} : Y \rightarrow \{0, 1\}$ that fits into this diagram.



But Y is connected and so $\bar{g}$ must be constant. Therefore $g = \bar{g}f$ is constant. $\square$

Theorem 1.1.10

Suppose $X = \bigcup_{\alpha \in A} X_\alpha$ and that for each $\alpha \in A$ the space X_α is (path) connected. If there is a point $x \in \bigcap_{\alpha \in A} X_\alpha$ then X is (path) connected.

Proof. 1. Connected: Let $g : X \rightarrow \{0, 1\}$, since X_α is connected, g is constant on X_α . Specially, $g|_{X_\alpha} \equiv g(x)$ for each $\alpha \in A$. Thus $g \equiv g(x)$ is constant. Thus X is connected.

2. Take any $y_1, y_2 \in X$, suppose that $y_1 \in X_{\alpha_1}, y_2 \in X_{\alpha_2}$. Since $x \in X_{\alpha_1} \cap X_{\alpha_2}$, there exists path γ_1 on X_{α_1} connecting y_1 and x , and path γ_2 on X_{α_2}

connecting x and y_2 . We define $\gamma : [0, 1] \rightarrow X$ to be

$$\gamma(t) = \begin{cases} \gamma_1(2t), & t \in \left[0, \frac{1}{2}\right] \\ \gamma_2(2t - 1), & t \in \left[\frac{1}{2}, 1\right] \end{cases}$$

Then it follows from the gluing lemma that γ is continuous, connecting y_1 and y_2 . Thus X is path-connected. □

1.2 Constructions and Connectedness

Theorem 1.2.1

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of (path) connected topological spaces. Then $X := \prod_{\alpha \in A} X_\alpha$ is (path) connected.

Proof. 1. Conncted: We only show that $X \times Y$ is conncted if X, Y are. The infinite case is complicated. Fix a point $b \in Y$, let

$$T_x = (X \times \{b\}) \cup (\{x\} \times Y)$$

is connected space since it is the intersection of two conncted space with common point (x, b) . Then

$$X \times Y = \bigcup_{x \in X} T_x$$

is connected since it is the intersection of connected spaces with the common point (a, b) .

2. Path-connected: For each $a, b \in X$, and each $\alpha \in A$. Since X_α is path connected, there exists a path $p_\alpha : [0, 1] \rightarrow X_\alpha$ connecting a_α and b_α . Then it follows from the universal property that there exists a unique continuous function $p : [0, 1] \rightarrow X$ such that $\pi_\alpha \circ p = p_\alpha$. Thus $p(0) = (a_\alpha)_{\alpha \in A} = a$, $p(1) = (b_\alpha)_{\alpha \in A} = b$. □

Example 1.2.2 ▶ A cartesian product of connected spaces that is not conncted in the box topology

Consider the $\mathbb{R}^\omega$ in the box topology. We write the set A the set of all bounded suquences of real numbers, and B of all unbounded sequences. If $\mathbf{a}$ is a point of $\mathbb{R}^\omega$, then open set

$$U = (a_1 - 1, a_1 + 1) \times \cdots$$

consists entirely of A or B .

1.3 Components and Local Connectedness

Definition 1.3.1

Given X , define an equivalence relation on X by setting $x \sim y$ if there is a connected subspace of X containing both x and y . The equivalence classes are called the **components** (or the “connected components”) of X .

Theorem 1.3.2

The components of X are **connected disjoint** subspaces of X whose union is X , such that **each** nonempty connected subspace of X intersects only one of them.

Definition 1.3.3

We define another equivalence relation on the space X by defining $x \sim y$ if there is a path in X from x to y . The equivalence classes are called the **path components** of X .

Definition 1.3.4

A space X is said to be **locally connected at x** if for every neighborhood U of x , there is a connected neighborhood V of x contained in U . If X is locally connected at each of its points, it is said simply to be **locally connected**. Similarly, a space X is said to be **locally path connected at x** if for every neighborhood U of x , there is a path-connected neighborhood V of x contained in U . If X is locally path connected at each of its points, then it is said to be **locally path connected**.

Theorem 1.3.5

A space X is locally connected **if and only if** for every open set U of X , **each component of U** is open in X .

Proof. Suppose X is locally connected. Take any open set U of X , and any component C of U . For each $x \in C$, there exists a connected neighborhood V of x such that $V \subseteq U$. Since $V \cap C \neq \emptyset$, we have $V \subseteq C$. C is open.

If for every open set U of X and each component C of U , C is open. Then for each $x \in X$, and each neighborhood V of x . The component C of V containing x is a open connected neighborhood of x contained in V . Thus X is connected. $\square$

Theorem 1.3.6

A space X is **locally path connected if and only if** for every open set U of X , each path component of U is open in X .

Theorem 1.3.7

If X is a topological space, each path component of X lies in a component of X . If X is locally path connected, then the components and the path components of X are the same.